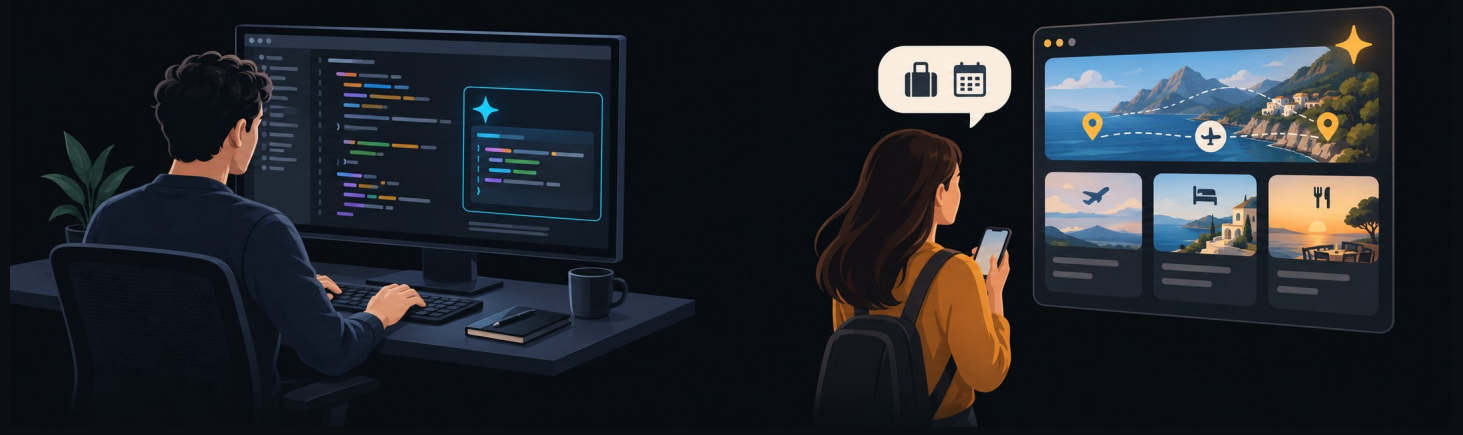
AI Changes the Delivery System and the Product Itself

PROBLEM → HYPOTHESIS → BUILD → VERIFY → REFRAME



HOW WE BUILD

My diagnosis: the SDLC is out of balance.
Rebalancing needs organisational change.

Small teams sharing context and roles
may coordinate more easily.

WHAT WE BUILD

Agentic projects need a different approach.
Models construct the solution at runtime.

Narrow, low-risk delegation may need lighter controls.
Rigor follows the possible consequences.

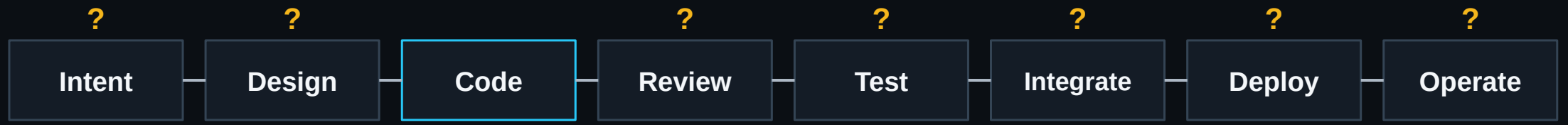
Vitalii Oborskyi Head of Delivery & PMO · Author of Uncertainty Architecture

DevCon Romania · 4 November 2026

vitaliioborskyi.ai

Constraints Across the Delivery System

Review automation can move the constraint to testing or integration.



Thought experiment: comparable changes/month; all admitted; fixed downstream capacity.

	Generated	Review capacity	Integration	BACKLOG
BEFORE Balanced flow	8	8	8	0 no accumulation
AFTER AI Other stages have not sped up	100	8	8	+92 waiting / month

ADMISSION SETS THE LOAD

Validated demand and user adoption also constrain value.

Limit admitted work: WIP stays bounded; excess demand waits outside.

The Comprehension Gap Limits Responsible Ownership

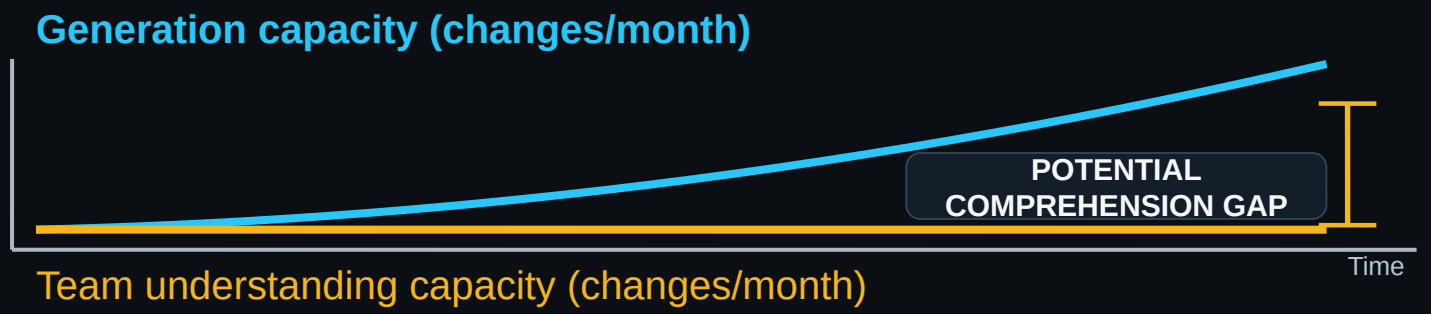
VERIFICATION

Evidence about the change

LEARNING FOR COMPETENT OWNERSHIP

- Knowledge transfer
- Shared understanding
- Competent ownership
- Operability & recovery
- Architecture & maintainability

Zoom in: CODE GENERATION ↔ REVIEW



WHAT IF NOT?

Can we generate faster?
YES

Can understanding keep pace?
NOT IN THIS SCENARIO

Can we explain, maintain and recover it?

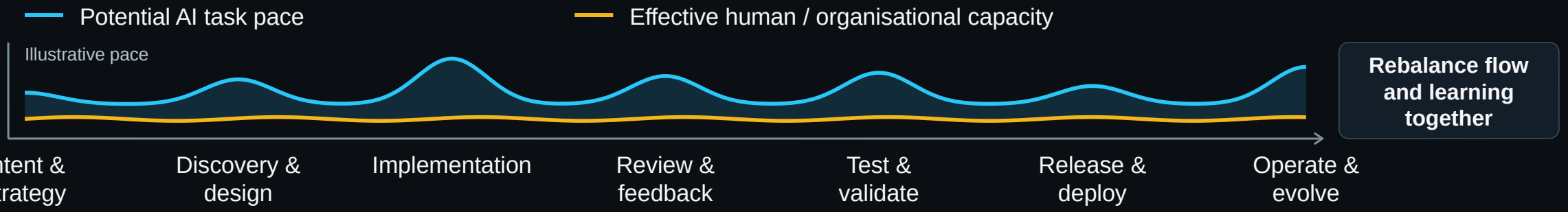
TWO FUNCTIONS Adequate checks and the shared understanding needed to act.
Assigning accountability alone does not supply that operational competence.

AI EXPERIMENT · 69 PARTICIPANTS · TASK COMPLETION ↑ / CODE UNDERSTANDING ↓

Martin-Lopez et al. · IEEE TSE 2026 · 14 professional developers.

Short tasks; individual understanding, not team ownership. Graph illustrative.

The same human learning constraint acts across the SDLC



Lifecycle illustration: no measured values, stage rankings or throughput estimates.

What Are We Actually Measuring?

Activity, delivery, stability and maintainability can move differently.

ACTIVITY · RELEASES · MARKETPLACE SIGNALS

NBER · DIFFERENT SAMPLES / TOOL STAGES · ESTIMATES VS BASELINE

25.5×	3.4×	1.3×
CHANGED LOC · ALL	COMMITS · ALL	RELEASES · NO ASYNC

MARKETPLACES · NEW SUPPLY VS EARLY USE

~2×	≈ FLAT	78→87%
NEW iOS APPS	EARLY RATINGS	APPS <10 RATINGS

Android downloads rose modestly; usage proxies differ.

CHEN–STRATTON · AGENTS · 1 NOV 2026 · 655 FIRMS

+35.7%	+13.7%	+56.1%
OPENED PRs	JIRA ISSUES RESOLVED*	REVIEW SUBMISSION TO APPROVAL

PR/Jira: per worker-month vs baseline. *Borderline; time ≠ reviewer hours.

SUPPORTING · DORA 2025 · SURVEY n=4,867

AI ADOPTION ↔ THROUGHPUT + INSTABILITY

Self-reported effects; survey association.

Instability = failed changes + unplanned fixes.

SUPPORTING · AGARWAL 2026 · 1,197 REPOS

+34.85% / +42.87%	+17.73% / +19.00%*
COGNITIVE COMPLEXITY · AF / IF	STATIC WARNINGS · AF / IF

AF: agent-first · IF: IDE-first · *not statistically significant.

Matched OSS cohorts. Proxies ≠ defects or cost.

MAINTAINABILITY · GITCLEAR 2026 · 623M CHANGES

MOVED CODE	−70%	2023→YTD26
DUPL. BLOCKS	+81%	2023→YTD26

Duplication per 1M changed lines; no causal claim.

No one productivity number. Measure useful outcomes, stability, maintainability and full cost.

Different samples/methods; no common funnel. Mechanism evidence ≠ measured industry prevalence.

A Viable Operating Model for This Context

Six hypotheses. Evidence decides where each is useful.

SPEED & FULL COST	DELIVERY & PRODUCT QUALITY	KNOWLEDGE & OWNERSHIP
Small batches + WIP limits Limit admitted work. Where does waiting move?	**Agentic review + test oracles** Scale checks with evidence. Shared blind spots? Who learns?	**Living architecture + docs** Explain decisions and changes. Who verifies it? Who learns?
Spec-driven development Durable intent and contracts. Do specs stay valid and useful?	**Disposable implementation** Regenerate bounded features. State and integration limits?	**Repository intelligence** Map dependencies and owners. Is the map grounded and current?

CONTEXT-SPECIFIC OPERATING MODELS Extensive delegation / limited assistance / no AI

Work, domain, complexity, criticality, legacy, team capability and economics

OWNER Authority, budget and cross-functional support

PILOT Baseline, comparable work, useful outcomes, full cost and stop conditions

SAFE RECOVERY DRILL Another engineer uses normal on-call tools and docs to explain, diagnose and restore. Business-defined time. Compare equivalent scenarios before and during the pilot.

When Model Judgment Enters Runtime

We implement the decision rules. The model constructs the solution at runtime.

EXPLICIT RULES / SIMPLIFIED PATH

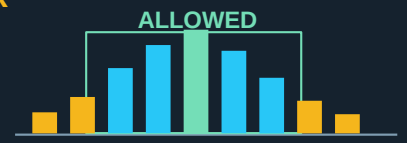
$$y = f(x)$$

Same relevant inputs + state + version

MODEL JUDGMENT / CONDITIONAL BEHAVIOR

$$y \sim P(y \mid x, c)$$

c = context + model configuration



ENGINEERING	WE SPECIFY AND IMPLEMENT THE RULES	WE BOUND THE MODEL'S CHOICE OF SOLUTION
REQUIREMENT	Specify intended behavior Implement the decision rules	Define acceptable paths and outcomes Set boundaries and stop conditions
BUG / TESTS	A rule violation can be exposed by one failing test	A wrong or harmful solution Test representative cases and repeated trials
DoR	Scope and requirements clear enough to implement	Risk, evaluation and control feasibility May be ready only for research
DoD	Increment meets agreed acceptance and quality criteria	Behavior evidence + working controls Tested fallback and operating limits

DEVOPS / MODEL UNCERTAINTY

Uncertainty also sits inside the application.
Monitor decisions, outcomes and controls.

NEW CLASS / THINKING SYSTEMS

Runtime judgment changes engineering and the project's risk profile.

Control effort follows delegation and consequences. Complex cases can require separate R&D.

Control Feasibility Before Delivery

Business scope and delegated authority define the control problem.

A WORKING DEMO

Shows that one path can work.

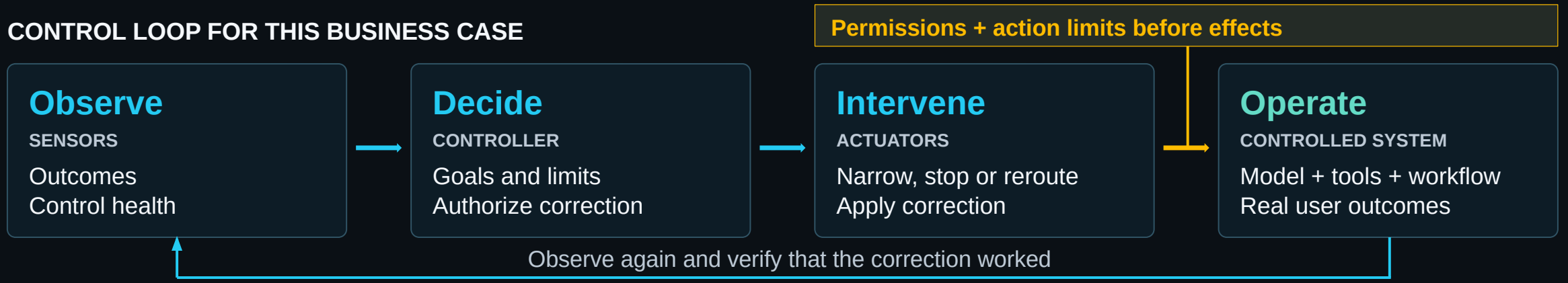
A few users leave many paths untested.

R&D FOR THE FULL CONTROL SYSTEM

Map failure modes, required expertise and operating cost.

Evidence before a production plan and budget.

CONTROL LOOP FOR THIS BUSINESS CASE



Human sensing or decisions make operator capacity an architectural dependency.

Availability, expertise, workload, response time and authority

RISK + CONTROLS

Failure modes and safe limits
Tested correction and recovery

EXPERTISE + FULL COST

Domain and control specialists
Staffed response and fallback

PROJECT DECISION

Proceed, narrow scope,
fund more R&D or stop

Control feasibility may require a separate research phase before delivery commitment.

Service Health and Decision Quality

A completed request can still contain an unsafe solution.

TWO ILLUSTRATIVE SAMPLES / 200 CASES EACH

REQUEST SUCCESS

98%

196 requests meet service criteria
4 technical failures

May meet the agreed service target

Availability, latency, load and error rate

SOLUTION QUALITY

98%

196 acceptable solutions
3 incorrect + 1 dangerous

One dangerous plan fails this safety gate

A plausible recovery plan can worsen an incident.

BEFORE RELEASE

Block release if the critical criterion fails.

DURING OPERATION

Stop affected automation. Use a safe fallback.

EVALUATE THE SOLUTION AND THE CAPACITY TO INTERVENE

SERVICE

Availability and latency
Errors and resource use
Operating cost

BEHAVIOR

Correct, useful solutions
Safe plans and actions
Outcome severity

CONTROLS

Detection and escalation
Time to intervene
Fallback effectiveness

PEOPLE / CAPACITY

Expertise and coverage
Workload + response time
Reported fatigue / overload

Human capacity is part of system health.

Set acceptance criteria and responses for the use case and consequences.

AI Changes Both Sides. The Engineering Must Follow.

Build the future we believe in, with sound engineering and readiness for what may go wrong.

HOW WE BUILD

Rebalance the SDLC.

FLOW · QUALITY · KNOWLEDGE

Measure outcomes,
recovery and cost.

THE SUBPRIME CODE CRISIS

Systems thinking · Theory of Constraints

WHAT WE BUILD

Engineer Thinking Systems.

BOUNDARIES · EVIDENCE · ACTION

Observe outcomes.
Retain authority to act.

UNCERTAINTY ARCHITECTURE

Systems thinking · Control Theory

BOTH PROJECTS · PUBLIC RESEARCH

https://vitaliioborskyi.ai/

Research · Writing · Contact

QUESTIONS?

DEVOPS / SRE	ENGINEERS / ARCHITECTS	CTO / DELIVERY / PRODUCT
Observe · intervene · recover	**Define bounds · test controls**	**Assign owners · fund evidence**